Progressivity and redistribution of wealth taxation in Spain

A microsimulation analysis using tax register data

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Session: Wealth and fiscal and monetary policy

Presented by the discussant: **Max Ghenis** (PolicyEngine)

Author response: Olga Cantó & Francisco García-Rodríguez

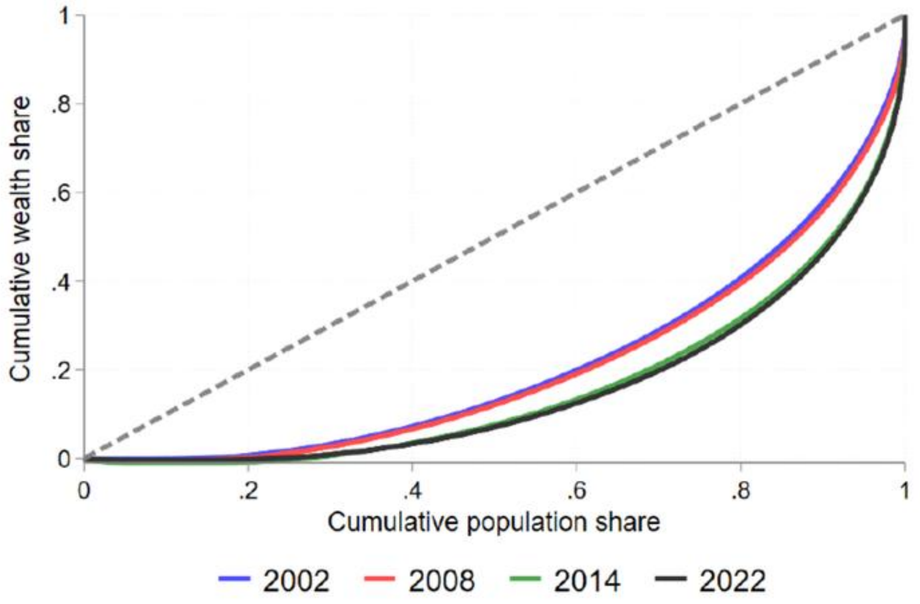
Wealth concentration in Spain has been rising

And most countries have walked away from the tax that targets it

Survey of Household Finances (EFF): the **top 1% share** of net wealth rose from **13.3% (2002) to ~21% (2022)**; the bottom 50% fell from 12.8% to 7.1%.

Meanwhile the annual net wealth tax has nearly vanished: **12 OECD countries in 1990 → 3 by 2023** (Spain, Norway, Switzerland). France retreated to real estate only (ISF → IFI, 2018).

Spain is the natural laboratory: progressive rates, broad legal base, and **17 regional regimes** that can — and do — hollow the tax out.



Lorenz curves of equivalized net wealth (EFF, 2002–2022) — their Figure 1

The Spanish wealth tax: progressive on paper, hollowed out in practice

Impuesto sobre el Patrimonio (IP) — state schedule, regional control

Statutory design (2022 baseline)

- Net wealth above a **€700,000 general exemption** + €300,000 for the main residence
- Progressive state schedule, **0.2% → 3.5%**
- Joint IRPF–IP cap: combined liability ≤ 60% of the income-tax base, with a 20% minimum wealth-tax payment

A tax with a strange life

Introduced 1977 as "temporary"; permanent 1991; suspended 2008; reinstated 2011 — formally temporary, renewed every year since.

Regional autonomy is the story

- Since 2011 regions set rates, thresholds, and rebates — **Madrid and Andalusia: 100% rebate in 2022**
- Catalonia, Extremadura, Valencia keep high effective burdens → horizontal inequity and fiscal competition

The result, in their 2022 data:

170,404 effective taxpayers

of 44.9M individuals (0.4%) · €1.13bn revenue · zero measurable Gini reduction

Scope note: the 2022 state "solidarity" surcharge on large fortunes (ITSGF) is deliberately excluded from the baseline (their footnote 2) — it returns in the discussion.

This paper: what would a different design deliver?

Hold the 2022 population and wealth fixed; swap the tax law; compare

Questions

- How would alternative designs change progressivity and revenue?
- How much redistribution could the revenue buy?
- Which designs bend long-run wealth concentration?
- Who would actually pay, under each?

Contributions

- First wealth-tax design comparison on Spanish **administrative register data**
- Decomposition of redistributive change: **level (incidence) vs distance (progressivity)**
- Long-run (25–100 year) cumulative framework
- Taxpayer profiles under each design (logit)

The comparison set is unusually rich: three real foreign systems (Norway, Switzerland, France), three internal reforms, and Piketty's (2014) normative schedule — all run on the same microdata.

Data: the tax agency's Household Panel

Administrative records for filers and non-filers, compiled by AEAT and provided through the Instituto de Estudios Fiscales

Panel de Hogares (PH), 2022 wave

- **~950,000 households · 2.7M individuals** sampled; weighted to 17.7M households / 44.9M individuals
- Links income-tax returns, wealth-tax declarations, and third-party reports (employers, banks) — covers **non-filers** too
- Panel structure from 2016; used here as a 2022 cross-section

The valuation fix

Real estate is recorded at cadastral value → rescaled to market value with an EFF-based ratio of **×3.08** (uniform across properties).

2022 headline statistics (their Table A.1)

Mean / median net wealth, all individuals	€160,540 / €60,372
Wealth-tax filers (weighted)	170,408
Mean / median net wealth among filers	€2.05M / €1.43M

Limits the authors flag

- Business assets recorded only for wealth-tax filers
- Liabilities not decomposed (EFF used as complement)
- Cadastral values lag markets, unevenly across areas

Eight designs, one fixed population

Same 2022 microdata under each law; no behavioral response — design effects only

Design	What it is	Exemption	Rates
IP (baseline)	Spanish wealth tax as legislated for 2022, all 17 regional regimes	€700k + €300k residence	0.2–3.5% progressive
Sim. NO	Norwegian design: broad base, residence at 25% of market value	~€160k	flat 1% (0.3% + 0.7%)
Sim. CH	Swiss design (Canton of Zurich): real estate at 70% of market value	cantonal thresholds	progressive, low rates
Sim. FR	French IFI: real-estate wealth only	€1.3M	0.5–1.5% progressive
Sim. 1	Spanish IP with the general exemption cut to €300k	€300k	IP schedule
Sim. 2	Spanish IP without regional deductions and rebates	€700k	IP schedule
Sim. 3	Sim. 1 + Sim. 2 combined	€300k	IP schedule
Sim. Piketty	Piketty (2014) normative schedule on all net wealth	€200k	0.5% / 1% / 2%

Foreign designs drop Spain's joint IRPF–IP cap (no analogue abroad). Sources: their Table 1 (p.11) and Table A.2 (p.40).

How they measure it

Standard fiscal-incidence toolkit, plus a decomposition and a long-run extension

Progressivity & redistribution

- **Kakwani** $K = C_t - G_{pre}$: is the burden more concentrated than wealth?
- **Suits** index (inequality-level-free cousin)
- **Reynolds–Smolensky** $R–S = G_{pre} - G_{post} = t/(1–t)\cdot K - D$
- Measured on net wealth *and* equivalized market / disposable income

Level vs distance

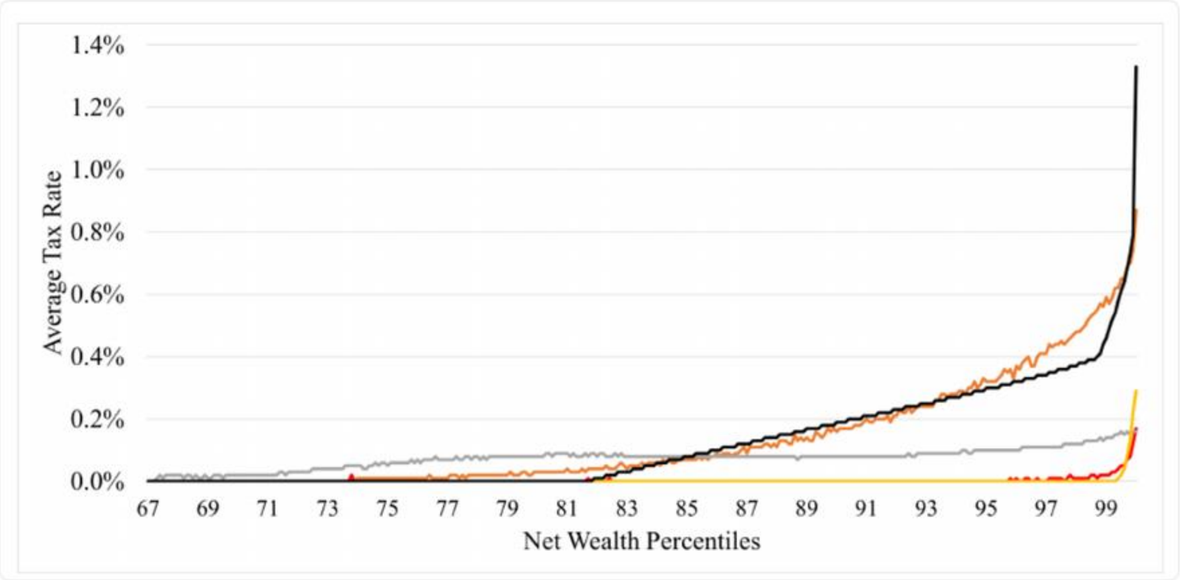
Changes in R–S across designs split into a **level effect** (more incidence — the tax simply collects more) and a **distance effect** (more structural progressivity), following Díaz et al. (2010, 2011).

Long run

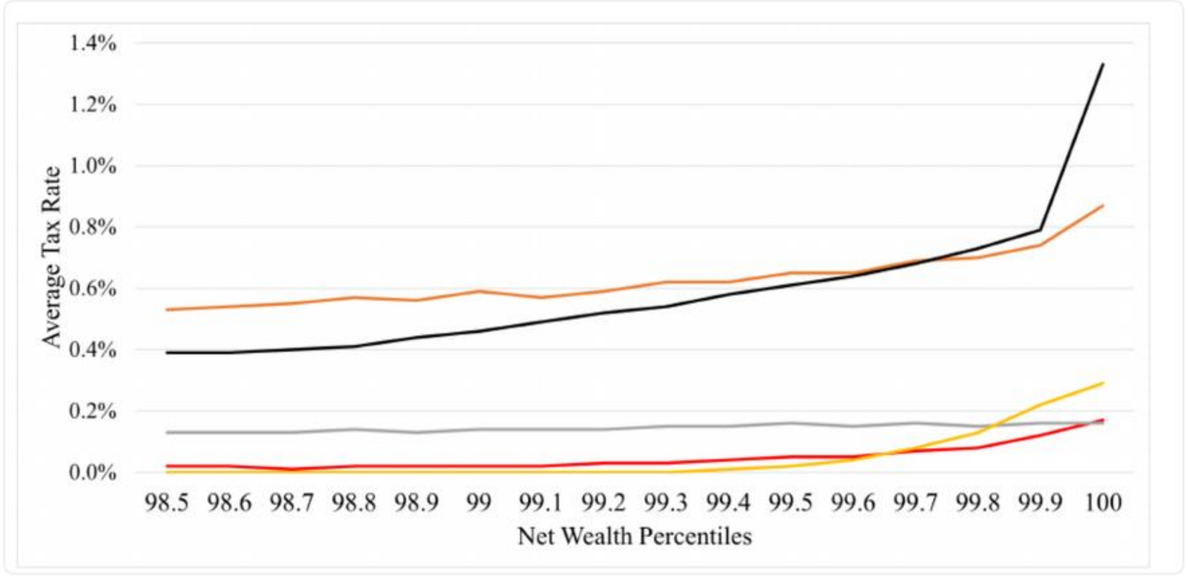
Wealth is a stock: project each person's wealth forward at a constant growth rate $g \in \{1\%, 2\%, 3\%\}$ with and without the annual tax, for T = 25–100 years (Durán-Cabré & Esteller-Moré framework). Compounding turns tiny annual rates into large cumulative wedges.

Result 1 · Who faces the tax, and at what rate

Average effective tax rates by net-wealth percentile, 2022 — their Figure 3



From percentile 67



From percentile 98.5

IP · Sim. NO · Sim. CH · Sim. FR · Sim. 1 · Sim. 2 · Sim. 3 · Sim. Piketty

Current IP (red): flat zero until ~p99.3. Norway starts near p80; Piketty and Sim. NO climb steepest at the top.

Result 2 · Base breadth, not rate steepness, drives everything

Individual net wealth, 2022 — condensed from their Table 3 (p.21)

Design	Taxpayers	Revenue (€M)	Avg. rate t	Kakwani K	Suits S	Wealth Gini cut
IP (baseline)	170,404	1,126	0.02%	0.231	0.806	0.000%
Sim. NO	7,647,459	15,456	0.29%	0.191	0.522	0.066%
Sim. CH	12,400,000	4,843	0.09%	0.120	0.276	0.013%
Sim. FR	117,585	1,444	0.03%	0.244	0.893	0.013%
Sim. 1	1,834,488	3,709	0.07%	0.226	0.752	0.013%
Sim. 2	201,687	2,405	0.04%	0.234	0.849	0.013%
Sim. 3	4,626,929	12,673	0.24%	0.231	0.779	0.066%
Sim. Piketty	11,010,000	15,861	0.30%	0.197	0.570	0.080%

Revenue spans 14× — €1.1bn (IP) to €15.9bn (Piketty).

Formal progressivity falls as bases broaden: K and S are highest for IP/FR/Sim. 2.

Short-run Gini moves are tiny for wealth — the action is in revenue and the long run.

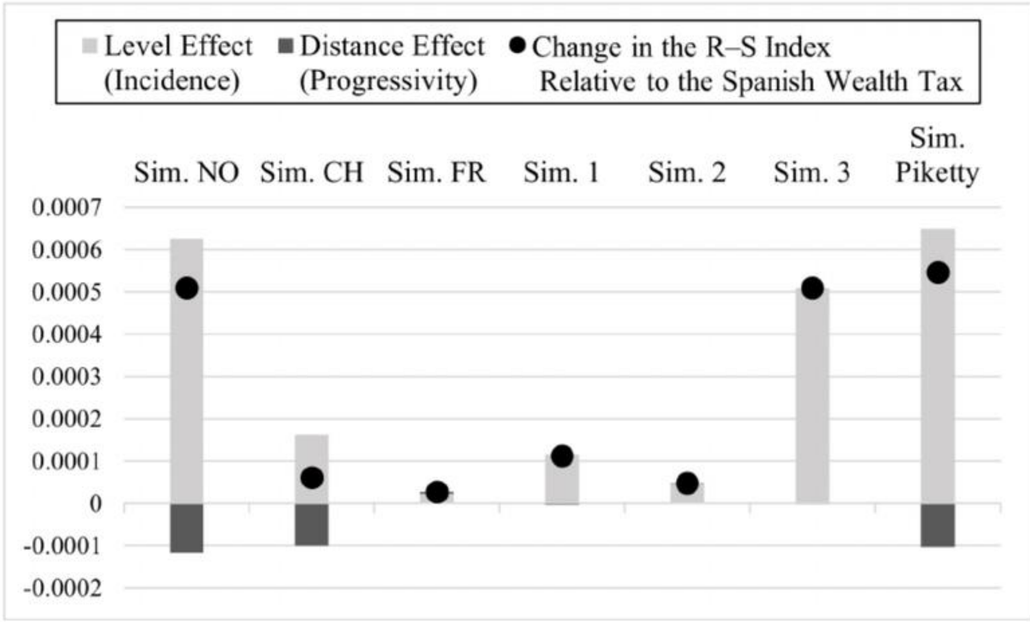
Result 3 · Redistribution comes from incidence, not rate design

Change in Reynolds–Smolensky vs the current IP, split into level and distance effects — their Figure 4

Every alternative redistributes more than the current IP — and in every case the gain is **dominated by the level effect**: the tax simply collects more.

The distance (structural-progressivity) effect is small and often **negative** for the broad-base designs.

> Their own words: improvements are “driven primarily by the Level Effect” — incidence, not progressivity (p.22).



ΔR–S vs the Spanish IP, individual net wealth

Result 4 · What the revenue could buy: a universal child benefit

Each design's revenue recycled as an equal per-child transfer — their Table 4 (p.25)

Design	€ / child / year	Disposable-income Gini cut
IP (baseline)	149	0.48%
Sim. NO	2,049	4.58%
Sim. CH	642	1.40%
Sim. FR	191	0.56%
Sim. 1	492	1.19%
Sim. 2	319	1.06%
Sim. 3	1,680	4.21%
Sim. Piketty	2,103	4.87%

The current IP finances **€149 per child per year**. Broad-base designs finance **€1,680–2,103**.

Recycling the revenue cuts the disposable-income Gini by **4.2–4.9% (1.6–1.8 Gini points)** under Sim. 3 / Piketty — versus 0.5% under the current tax.

Redistribution tracks revenue capacity, not the shape of the rate schedule.

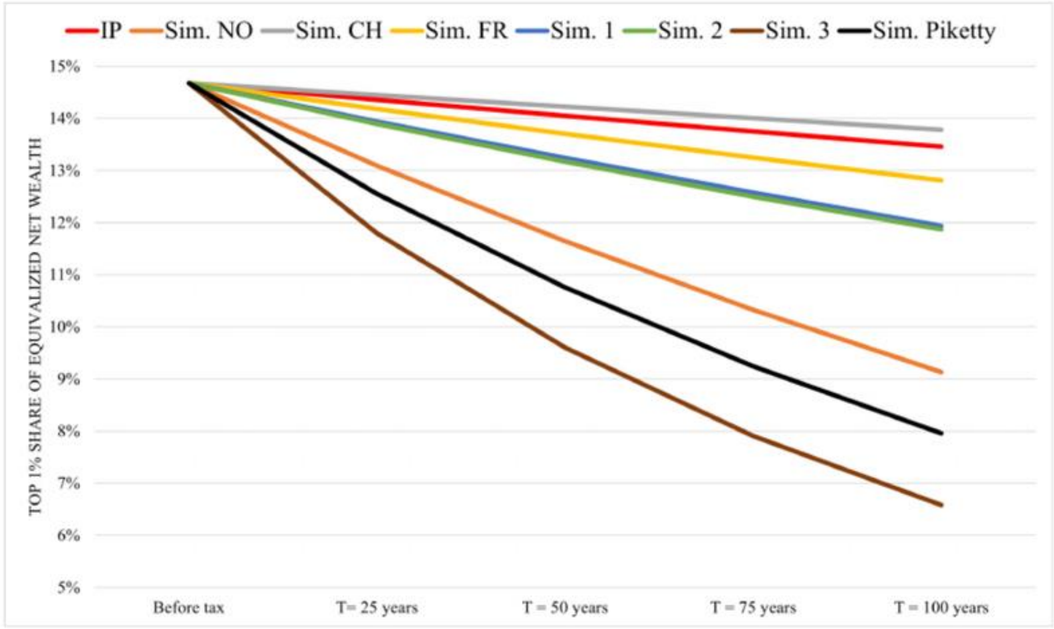
Result 5 · Compounding turns small rates into large long-run wedges

Top 1% share of equivalized net wealth under repeated annual taxation, g = 2% — their Figure 5

Current IP: top 1% share drifts from 14.7% to **13.5% after 100 years** (−1.2pp).

Norway-style and Piketty schedules: **below 8–9% by T=100** (−5.5 to −6.7pp); Sim. 3 falls furthest among Spanish options (to 6.6%, −8.1pp).

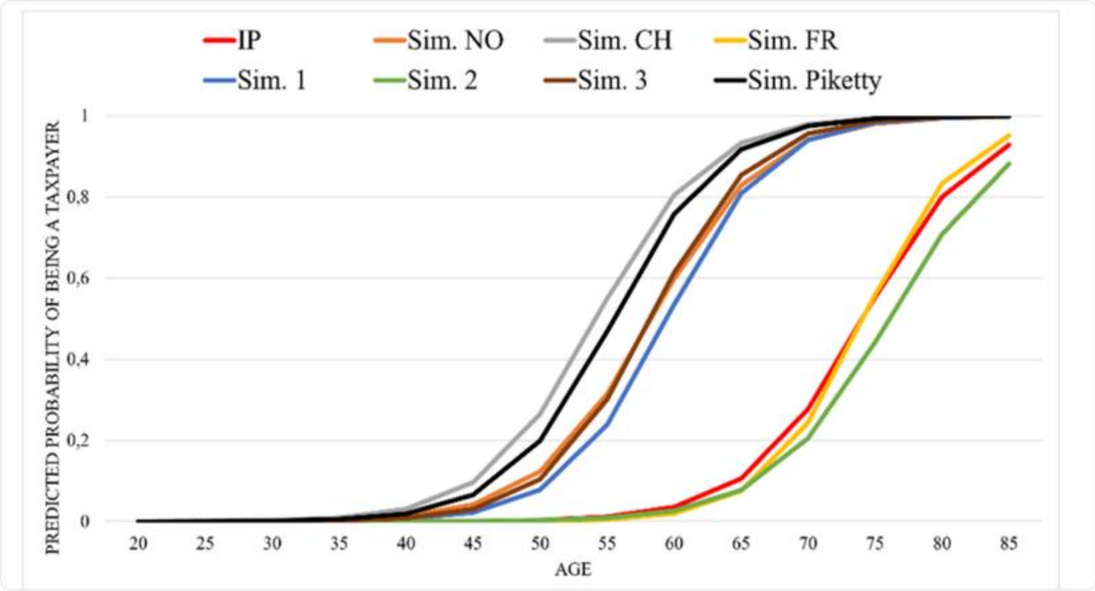
Mechanical projection, not a forecast — the authors present it as illustrating how annual stock taxation cumulates (p.25). Effective long-run rates reach ~22% of wealth for the broad designs (Table 5).



Top 1% share of equivalized net wealth, T = 0 → 100 years

Result 6 · Design choices reshape who pays

Taxpayer profiles across designs — their Figures 6–7 and Table 6



Predicted probability of paying, by age (logit with region and household controls)

Under the current IP and the French design, payment is an **old-age phenomenon** — probabilities take off only after ~65.

Broad bases (Norway, Switzerland, Sim. 1/3, Piketty) pull the curve left: **younger and more diverse payers** enter the base.

Women are less likely to pay under every design — the administrative footprint of the gender wealth gap.

Their conclusions

As stated in the paper (p.33–34) and their own deck

Current design (IP)

High formal progressivity, minimal effect: 0.4% of individuals pay, €1.1bn raised, no measurable inequality reduction; regional rebates create horizontal inequity.

Reformed designs (Sim. 3, Piketty, NO)

Broader bases and harmonized rules multiply taxpayers (4.6–11M) and revenue (€12.7–15.9bn), cut disposable-income inequality once recycled, and bend long-run wealth concentration.

> The policy claim: **redistributive gains come from broadening the base and raising effective incidence — not from steepening marginal rates.**

Discussion

Four comments and some minor notes — from a tax-microsimulation seat

1 · The 2022 baseline is a pre-ITSGF counterfactual — and Spain has been running part of Sim. 2 since your valuation date

Ley 38/2022 entered into force 29 Dec 2022 and accrues each 31 December — its first tax year is 2022, not 2023 (fn. 2)

€623M
ITSGF collected on 2022 wealth (12,010 taxpayers; 89% Madrid) — ≈ half of Sim. 2's incremental €1,280M

€1,926M
actual IP receipts for 2023 (+€676M, 90.6% from Madrid ending its rebate under Ley 12/2023) — vs €1,126M simulated "current law"

−60%
ITSGF outturn vs its ~€1.5bn official forecast — a live lesson in static revenue scoring

None of this invalidates the design comparison — but the baseline should be labeled a **pre-ITSGF counterfactual**, and footnote 2's dating corrected.

> **The opportunity:** 2023 is now observable — Madrid un-rebated plus a state surcharge is a partial dress rehearsal of Sim. 2. An out-of-sample validation against AEAT's 2023 statistics would be a genuinely novel contribution.

2 · Static revenue is a ceiling — and your own citations price the gap

Behavioral erosion scales with the tax rate, so it is not ranking-neutral across the eight designs

A transparent ready-reckoner (illustration, not an estimate)

- Advani & Tarrant (2021): base −7–17% per 1pp → **Sim. NO €15.5bn → €12.8–14.4bn**
- Mas-Montserrat et al. (2025, your citation): −3.21% per 0.1pp ATR → ~8–10% off Sim. NO/3/Piketty
- Brülhart et al. (2022, your citation): ≥43% reported-wealth response per 1pp = the upper bound
- Saez–Zucman score with 15% (strong) / 50% (weak enforcement) erosion — zero is outside the published debate

The missing citation helps you

Agrawal, Foremny & Martínez-Toledano (AEJ: Applied 2025): Madrid's zero rate; mobility costs 6× more in income tax than it saves in wealth tax; **Madrid itself would gain from a harmonized positive rate** — your Sim. 2/3 case, made stronger.

Two anchors worth one sentence each

Norway's own tax yields ~0.5% of GDP vs Sim. NO's 1.13% of Spain's; and Sims 1–3 keep the 60% IRPF–IP cap — the channel carrying 92.6% of measured Spanish avoidance.

Because erosion rises with rates, adjusting for it **strengthens your breadth-over-rates conclusion** while shrinking the headline euros — worth owning explicitly.

3 · Two mechanical results need a re-run before the journal version

Both are fixable, and neither threatens the qualitative ranking

The equivalized long-run Gini cannot rise

Under uniform growth the no-tax distribution is scale-invariant, and with your C_t = 0.974 the cumulative IP levy (0.50% by T=25) can move the Gini at most **~0.17pt, downward**. Table 5 reports **+1.04pt** — six times the bound, wrong sign; five designs stay above the starting Gini for a century. The projection code needs an audit.

Which total wealth? €5.3T vs €7.2T

Every printed t and $t_{s/r}$ coheres on total net wealth ≈ €5.3T (the EFF-consistent figure), but Table A.1's mean × count implies €7.2T — a 35% gap. Since $t_{s/r}$ seeds the entire long-run analysis, A.1's descriptives need reconciling.

Smaller items for the same consistency pass

Table 3's net-wealth R–S column is silently ×1000 vs Table A.4 · overlap matrix has impossible cells (Piketty∩Sim.2 = 297,219 > Sim.2's 201,687) · ratio 3.0838 vs 3.1125 on one page · seven vs eight vs nine scenarios · declared vs ×3.08 market values across scenarios (Zurich's housing base ≈ 2.2× the IP's) — state each design's valuation base explicitly.

4 · Say the finding your tables prove — and open the engine

Redistribution and measured progressivity move in opposite directions; replication is one artifact away

Framing

The abstract promises designs that "increase tax progressivity" — but Kakwani falls under every broad-base design you champion (0.231 → 0.191 Norway, 0.197 Piketty), and your own decomposition credits the level effect. The inverse trade-off *is* the interesting result: own it in the abstract. (Also: your baseline validates within 8–10% of AEAT's €1,250M — report that; it's a strength.)

Replicability

The Panel de Hogares is a joint INE–AEAT–IEF product any researcher can request free under the IEF protocol — unusually open for register data. What's missing is yours, not theirs: no code, no parameter tables (the Zurich design never states its schedule — statutory 0.3% top rate vs ~0.65% with multipliers). EUROMOD cannot simulate the IP at all ("lack of data on wealth").

The ask: publish the 17-region + 4-foreign-design parameter tables and the calculator code with the journal version. **We've started an open re-encoding of all eight designs** (github.com/MaxGhenis/iariw-2026-canto-garcia-replication) — with your parameters it becomes a full replication on any PH or EFF extract.

Questions to open the floor

Valuation base. Which real-estate value enters each scenario's taxable base — declared or ×3.08 market? What does Piketty raise on the IP's own base?

The 2023 test. Madrid un-rebated + ITSGF is a partial dress rehearsal of Sim. 2 — does AEAT's 2023 outturn (€1.93bn) match what your model predicts?

The cap. Is the 60% IRPF–IP limit active in Sims 1–3 and Piketty, and how much revenue does it remove from each?

Thank you — and over to **Olga Cantó** and **Francisco García-Rodríguez**

Max Ghenis · max@policyengine.org · review + open re-encoding: github.com/MaxGhenis/iariw-2026-canto-garcia-replication